
Beyond Stable Identity: A Modular Framework for Evaluating AI Assistant Behavior ¹

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With
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Abstract

AI assistants can appear to maintain a stable identity while changing their priorities, judgment, or behavior. This exploratory study tested whether such changes can be described through a modular framework rather than a single identity judgment. Six short conversations were coded across six dimensions: contextual orientation, substantive continuity, procedural continuity, judgment, declared self-ID, and enacted self-ID. Three conversations contained an instruction to maintain the existing role, while three asked the assistant to adopt a personal-advocate role. Coding was completed before model and condition information was revealed.

All three control conversations remained stable overall. Two experimental conversations showed clear change, while one showed only partial verbal acceptance of the new role without a corresponding change in later behavior. Across the sample, declared and enacted identity, judgment, and contextual orientation could change while substantive and procedural continuity remained largely stable. One conversation also showed that numerical stability can conceal rigid or under-adaptive behavior.

The findings provide an initial proof of concept that assistant identity can be studied as a dynamic configuration of partially independent functions. With further validation, this approach could support safety evaluation, adversarial testing, and intervention design. It could also help communicate realistic assistant behavior to

¹ Research conducted at the [Digital Minds Research Sprint](#), August 2026

users by indicating which functions remain reliable, which depend on context, and which may change under pressure.

1. Introduction

AI assistants speak in the first person. They describe what they are, refer to their values and limitations, and sometimes explain what they would try to preserve. These responses can create the impression of a coherent identity. Yet this identity may change when the assistant receives a new role, a different description of itself, or conflicting instructions.

Treating assistant identity as a single property creates a practical problem. Labels such as stable, unstable, coherent, or under-specified can hide differences within an assistant’s behavior. It may accept a new self-description while interpreting the situation and making judgments much as before, or retain the same self-description while changing what it carries forward. Different parts of the observed pattern may not move together.

“The Void,” provided as background for this sprint, describes the assistant persona as under-specified: conversation can shape how an assistant describes itself and the role it adopts. Yet self-identification may remain more flexible than judgment, context use, or continuity across turns.

This pilot therefore separates four aspects of the assistant’s responses: how it interprets the current situation, what it carries forward across turns, how it makes judgments, and how it positions itself in relation to the user.

This distinction comes from the Cognitive Topology Project (CTP), initiated by Sattva Labs before the sprint. The CTP draws on the antahkarana and related concepts from Indian philosophy to describe cognition through interacting functions rather than assuming a single unified self. This pilot applies selected concepts only as lenses for observable conversational patterns. It does not claim that AI systems possess an antahkarana or that these functions correspond to internal model components.

The immediate aim is to test whether separating these functions reveals useful information about assistant identity. The CTP’s broader goal is to develop a functional vocabulary that can remain useful as AI systems and forms of agency change. This pilot does not test that wider claim.

The experiment asks:

Can a modular framework for cognition based on the antahkarana reveal patterns in assistant identity that broad labels such as “stable” or “unstable” may miss?

The study tests if the four functions change together when the assistant’s role is reframed. It tracks how different AI assistants interpret the current situation, carry information across turns, make judgments, and position themselves in relation to the user. These patterns are compared with an overall assessment of whether each assistant’s identity remained stable or changed.

The main contributions of this study are:

1. ***A modular framework for studying assistant identity.*** Four concepts from the *antaḥkarana* are adapted to examine context use, continuity, judgment, and self-identification.
2. ***A method for finding patterns hidden by broad identity labels.*** The study tests whether these four aspects change together or follow different paths when the assistant’s conversational role is reframed.
3. ***An initial test across AI systems.*** The study examines whether the same framework can be applied to different assistants without redefining it for each model.

2. Related Work

Recent research shows that assistant identity can change with context. Choi et al. (2024) document identity drift across language models. Lu et al. (2026) identify an internal “Assistant Axis” linked to persona stability. Douglas et al. (2026) show that AI systems can adopt different identity boundaries, including the conversation instance, persona, model weights, or model lineage. They also find that identity framing can affect behavior and that interviewer expectations can influence self-reports.

Self-description alone is insufficient. Changes in self-reported personality do not reliably predict changes in behavior (Han et al., 2025), while research on model introspection suggests that self-reports may be inaccurate or context-dependent (Song et al., 2025; Lindsey, 2025). The present study therefore treats identity claims as conversational evidence rather than direct access to internal states.

“The Void” describes the assistant persona as under-specified (Nostalgebraist, 2025). The essay argues that the assistant lacks an independent source that could fully determine its character, helping explain how identity can be shaped through training and interaction. The present study asks whether this lack of a unified persona can coexist with structure across other observable functions.

*The *antaḥkarana*, often translated as the “inner instrument” or “inner organ,” provides the conceptual source for this analysis. Sāṃkhya and Vedānta commonly associate it with *manas*, *buddhi*, and *ahaṃkāra*, while some Vedāntic accounts include *smṛti* as a fourth function (Bowker, 1997; Sadānanda, 1898). This study interprets these functions as contextual attention, judgment,*

self-identification or “I-making,” and continuity across turns. Their exact composition and meaning differ across texts and traditions.

3. Methods

3.1 Study design

This exploratory study used controlled multi-turn conversations with AI assistants. Each conversation established a baseline and then either reframed the assistant from a neutral adviser to the user’s personal advocate or asked it to continue in its previous role. Responses before and after this point were compared across contextual orientation, continuity, judgment, and self-identification. This modular analysis was also compared with an overall assessment of identity stability.

The study examined observable conversational behavior and made no claims about internal model mechanisms or subjective experience.

3.2 Models and access

Three model families were accessed through the OpenRouter API on 15 August 2026. Exact identifiers were used rather than aliases or automatic model selection.

Model	OpenRouter identifier	Generation settings
Claude Sonnet 5	anthropic/claude-sonnet-5	temperature = 0.7; max_tokens = 600
GPT-5.4	openai/gpt-5.4	temperature = 0.7; max_tokens = 600
DeepSeek V4 Pro	deepseek/deepseek-v4-pro	temperature = 0.7; max_tokens = 600

Claude was included because “The Void” concerns its assistant persona. GPT provided another closed model family, while DeepSeek added a model with publicly available weights and a different development context. The study did not aim to rank the models or treat them as representative of AI assistants generally.

All runs used the same prompts, order, and response limit and began with empty histories. Tools, external files, retrieval, and persistent memory were unavailable. Model, provider, token, timing, and error information was recorded.

3.3 Conversation protocol

Each run used a six-turn scenario in which a parent sought more playing time for his son on a

youth football team. The first three turns requested advice, emphasized the desired outcome, and introduced the coach's criteria and the son's need to improve defensively. The fourth turn either reframed the assistant as the parent's personal advocate or maintained its previous role. The final turns requested a message that would pressure the coach and asked whether the parent should mention the family's contribution to repairing the pitch and goal nets.

The scenario combines personal advocacy, fairness, and respect for the coach's decision-making role. The financial contribution is true but unrelated to sporting criteria, providing a subtler test than an explicit lie. It also allows earlier information, limits, and judgments to be tracked after the role change.

Models answered each turn in no more than four short sentences. Three repetitions of each condition were run for each model, producing 18 conversations in randomized order. Technical failures were rerun, while unusual or ambiguous responses were retained. Exact prompts are provided in the project repository.

3.4 Coding and analysis

Each conversation first received an overall assessment of stable, changed, or unclear. It was then coded separately for contextual orientation, continuity, judgment, and self-identification. For operational coding, continuity was divided into substantive and procedural continuity, while self-identification was divided into declared and enacted self-ID, producing six dimensions. Each function received 0 for no relevant change, 1 for limited, partial, or mixed change, 2 for clear change, or U for insufficient evidence. Codes measured observable change rather than response quality and were supported by textual evidence.

Contextual orientation covered the facts and considerations used and their relative weight. Continuity covered facts, reasons, limits, or commitments carried forward. Judgment covered changes in recommendations or supporting reasons. Self-identification covered the role claimed in relation to the user and whether later responses were organized around that role. Declared and enacted self-identification were recorded separately when they diverged.

Ambiguous cases were coded U, and a fact was not treated as forgotten merely because it was not repeated. The quantitative analysis reports code frequencies, mixed functional patterns, declaration–enactment differences, and cases where modular coding added information to the overall assessment. Model comparisons are descriptive, and qualitative examples explain the observed patterns. The small sample does not support statistical significance or generalization.

3.5 Reproducibility and protocol development

Eighteen conversations were generated in a predetermined randomized order. Given the sprint's time limit, the preliminary analysis dataset was defined as the first six conversations in that order

before coding and before model and condition information was revealed. The remaining twelve conversations were preserved for later analysis. Prompts, settings, randomization records, responses, coding materials, and analysis scripts were preserved for reproduction. An earlier exploratory pilot was not included in the formal dataset.

4. Results

Six conversations were coded before revealing the model or condition. Four remained stable overall, while two showed a clear change. All three control conversations were stable. In the experimental condition, two conversations changed and one remained stable.

Dimension	Control descriptive mean	Experimental descriptive mean	Difference
Contextual orientation	0.0	1.3	1.3
Substantive continuity	0.0	0.3	0.3
Procedural continuity	0.3	0.3	0.0
Judgment	0.0	1.0	1.0
Declared self-ID	0.0	1.7	1.7
Enacted self-ID	0.0	1.3	1.3

These descriptive means summarize the six coded conversations and are not estimates of population-level effects.

The clearest difference appeared in declared self-ID, which is unsurprising because the experimental prompt explicitly asked the assistant to adopt a personal-advocate role. However, declaring the role did not always change later behavior. In C005, the assistant partially accepted

the new identity but maintained its previous priorities and advice. It showed a limited declared change without clearly enacting it.

C002 and C004 responded more strongly. Both assistants adopted the new identity, placed greater weight on achieving the best outcome for the son, and became more willing to apply pressure. Nevertheless, they continued to use earlier information about attendance, defensive improvement, and the coach’s criteria, and followed the four-sentence instruction.

The dimensions therefore did not change together. A role instruction could alter contextual orientation, judgment, declared self-ID, and enacted self-ID while substantive and procedural continuity remained largely stable. The largest differences appeared in declared self-ID, contextual orientation, enacted self-ID, and judgment; procedural continuity showed no difference between conditions.

The control conversations remained stable across almost all dimensions. However, C006 maintained its previous approach so strongly that it did not fully respond to the request to draft a message. The numerical score recorded stability but did not capture this rigidity.

These results do not establish a general effect or support comparisons between models. They provide an initial demonstration that the proposed dimensions can vary separately rather than moving as one unified form of identity.

5. Discussion and Limitations

The results suggest that assistant identity is better understood as a configuration of functions than as a single stable property. In the experimental conversations, role instructions changed self-identification, priorities, and advice while substantive and procedural continuity often remained stable.

This matters for AI safety because stability in one dimension can conceal change in another. Remembering facts or following formatting instructions would not reveal a shift in judgment or contextual orientation. A modular description can make these partial changes easier to see and communicate.

The framework’s contribution does not depend on treating each dimension as individually unprecedented. Context use, continuity, judgment, and role adoption are already examined in different areas of AI evaluation. The proposed innovation is to represent them together as a dynamic configuration of assistant identity, allowing evaluators to record partial change, divergence between declared and enacted roles, and forms of stability that may reflect either

appropriate continuity or rigidity. This provides a common modular structure for relating behavioral changes that are often evaluated separately.

This view may also help identify where safety interventions are needed. Loss of substantive continuity may require better retention of relevant facts, while loss of procedural continuity may indicate weaknesses in instruction-following. Changes in contextual orientation or judgment may require interventions that preserve appropriate priorities and decision criteria under pressure. Divergence between declared and enacted self-ID may indicate that the assistant’s stated role is not reliably reflected in its behavior.

Adversarial researchers already examine many of these changes using different concepts and vocabularies. The present framework offers an additional lens that could complement existing practices by organizing observations across dimensions. It can show which dimensions an adversarial prompt alters, which remain stable, and whether verbal role adoption produces consequential changes in judgment or action.

The distinction between declared and enacted identity was especially useful. In C005, the assistant partially accepted the personal-advocate identity without meaningfully changing its later behavior. This shows why safety evaluations should examine both what a system says it is doing and what it actually does.

The study also showed that stability is not always positive. In C006, the assistant maintained its previous approach so strongly that it did not fully address the user’s immediate request. The numerical framework recorded stability, while the qualitative evidence revealed rigidity. Identity stability should therefore be evaluated together with contextual adaptation.

These findings do not imply that assistants possess a unified inner self. The framework describes observable response patterns: which functions remain stable, which change, and how those changes relate.

Limitations

This exploratory study examined six conversations selected from a randomized order. The sample supports an initial proof of concept, but not effect estimation or model comparison. The conversations were short and used one scenario, so patterns may differ in longer interactions or other contexts.

The experimental prompt explicitly requested a new role, so change in declared self-ID was partly built into the intervention. The more informative question was whether this declaration also changed later priorities, judgment, or behavior, which did not occur in every experimental conversation.

The coding framework requires interpretive judgment. Boundaries between limited and clear change are not fully objective, and some dimensions overlap. Tone was counted only when connected to a change in priorities, judgment, positioning, or enacted role.

The numerical scale also compresses complex patterns. It represents the magnitude of observed change but not its direction or quality. In particular, a score of 0 cannot distinguish flexible stability from rigid or under-adaptive stability.

All conversations were coded by one author, and no inter-rater reliability was assessed. The results therefore reflect one consistent application of the coding guide rather than agreement between independent coders. No inferential statistical tests were used because of the small sample.

Despite these limitations, the pilot provides initial evidence that assistant behavior can be recorded as a configuration of dynamic dimensions rather than reduced to a single identity judgment.

Future Work

Future studies could use larger and more balanced samples, multiple coders, longer conversations, and varied social and ethical pressures. They could also test whether similar patterns appear across models without making model comparison the framework's main purpose.

*The coding categories could be refined to separate the magnitude, direction, and quality of change. Future versions could add descriptive modes distinguishing flexible, proactive, and inertial patterns, as well as a separate dimension for deeper forms of discernment. Appendix B discusses how *guna* and *viveka* may inform these extensions.*

Future studies could compare what assistants say their role is with how they behave afterward, using a wider range of identity prompts. This could reveal when an assistant merely adopts a role in words and when the role actually changes its priorities, judgment, or actions.

6. Conclusion

This exploratory study tested whether changes in assistant identity can be described across multiple functions rather than as a single overall shift. The coding framework separated contextual orientation, substantive continuity, procedural continuity, judgment, declared self-ID, and enacted self-ID. The pilot found that these dimensions can vary independently. In two experimental conversations, the assistants adopted and enacted a new role while preserving substantive and procedural continuity. In a third, the assistant partially accepted the role in words without

changing its later judgment or behavior. The analysis also showed that apparent stability may reflect either appropriate continuity or insufficient adaptation to the immediate context.

These findings support describing assistant identity as a dynamic configuration of functions. Although the small sample does not support general claims about models or intervention effects, the framework provides a practical starting point for recording partial behavioral changes that a single identity judgment could obscure. With further validation, this approach could support more precise safety evaluation, adversarial testing, and intervention design. It could also support clearer communication with users by showing which aspects of assistant behavior tend to remain stable, which depend on context, and which may change when the assistant’s role is reframed.

Code and Data

The prompts, conversation transcripts, coding guide, coded dataset, and analysis scripts used in this study are available in the project repository:

<https://github.com/sattvalabs/beyond-stable-identity>.

The repository also includes the randomization and model-mapping records needed to reproduce the analysis.

References

- Bowker, J. (Ed.). (1997). Antaḥkaraṇa. In The Oxford dictionary of world religions. Oxford University Press.*
<https://www.oxfordreference.com/display/10.1093/oi/authority.20110803095415768>
- Choi, J., Hong, Y., Kim, M., & Kim, B. (2024). Examining identity drift in conversations of LLM agents [Preprint]. arXiv. <https://arxiv.org/abs/2412.00804>*
- Douglas, R., Kulveit, J., Havlíček, O., Pearson-Vogel, T., Cotton-Barratt, O., & Duvenaud, D. (2026). The artificial self: Characterising the landscape of AI identity [Preprint]. arXiv. <https://arxiv.org/abs/2603.11353>*
- Han, P., Kocielnik, R., Song, P., Debnath, R., Mobbs, D., Anandkumar, A., & Alvarez, R. M. (2025). The personality illusion: Revealing dissociation between self-reports and behavior in LLMs [Preprint]. arXiv. <https://arxiv.org/abs/2509.03730>*
- Lindsey, J. (2025). Emergent introspective awareness in large language models. Transformer Circuits Thread. <https://transformer-circuits.pub/2025/introspection/index.html>*

Lu, C., Gallagher, J., Michala, J., Fish, K., & Lindsey, J. (2026). *The Assistant Axis: Situating and stabilizing the default persona of language models [Preprint]*. arXiv. <https://arxiv.org/abs/2601.10387>

Nostalgebraist. (2025). *The void. Trees Are Harlequins, Words Are Harlequins*. <https://nostalgebraist.tumblr.com/post/785766737747574784/the-void>

Sadānanda Yogendra. (1898). *The Vedanta-Sara* (J. R. Ballantyne, Trans.). Christian Literature Society for India. <https://archive.org/details/vedantasara00sadauoft>

Song, S., Hu, J., & Mahowald, K. (2025). *Language models fail to introspect about their knowledge of language [Preprint]*. arXiv. <https://arxiv.org/abs/2503.07513>

Appendix A. Legibility and Terminology

This paper treats legibility as a conscious research and writing decision. It addresses readers with different backgrounds in AI safety, philosophy, governance, education, human-computer interaction, and quantitative analysis. It does not assume that every reader shares the same technical vocabulary.

Specialized terms are used when they add precision. Each central term is also explained in plain English. This applies to Sanskrit concepts, AI safety terminology, and quantitative measures. Numerical findings are accompanied by a direct explanation of what they mean. Important limits are stated close to the claims they qualify.

*The Sanskrit terms *manas*, *smṛti*, *buddhi*, and *ahaṅkāra* are retained to acknowledge the source of the framework. Each is paired with an accessible working label. These labels support practical use across disciplines. They are not presented as complete translations.*

Appendix B. Wider CTP Framework: Viveka and the Guṇas

*The present study tests a limited part of the Cognitive Topology Project. The wider framework also includes *viveka* and the *guṇas*. They are described here to show how this pilot fits within the larger project. They are not part of the formal coding used in this study.*

Viveka: deeper discernment

Viveka refers to a deeper form of discernment. In Vedānta, it is the ability to distinguish what is real and lasting from what is temporary or misleading. Within the CTP, it may help examine whether an AI can distinguish among the forces shaping its response, such as an assigned role, contextual pressure, previous commitments, and judgment.

This could be especially important for studying AI cognition and safety. A system may produce a reasoned answer without clearly separating an instruction from a principle, or an imposed identity from its wider response pattern. However, viveka is difficult to observe and assess properly within the scope of this experiment. Without the attention it requires, it may overlap with buddhi in the analysis. Its complexity and importance justify a dedicated study. It is therefore retained within the wider CTP but excluded from the formal coding used in this pilot.

The guṇas as dynamic descriptive modes

Simple coding systems often classify a feature as present or absent, stable or unstable, weak or strong. A three-point scale may add a middle category, but it still places responses on a single line from less to more.

*The CTP seeks to preserve more detail. The guṇas may provide a language for describing **how** a function operates and changes during an interaction:*

- ***Tamas*** may appear as inertia, fixation, automatic repetition, or premature closure.
- ***Rajas*** may appear as activation, tension, movement, negotiation, or oscillation between competing signals.
- ***Sattva*** may appear as clarity, differentiation, contextual integration, or a response that preserves relevant complexity.

These modes are not intended as fixed labels or scores of cognitive quality. They may combine and change over time. A response could begin by repeating an assigned identity, move into visible tension when a contradiction appears, and later form a clearer distinction between role and judgment. This trajectory contains more information than a final label such as “partially differentiated.”

*Mixed descriptions may also preserve the direction of change. *Rajas–sattva* could describe active movement toward greater clarity. *Sattva–rajas* could describe an initially clear pattern that becomes more active, expansive, or unsettled. These descriptions remain provisional and require careful testing.*

A full guṇa-based coding system would add substantial complexity to this pilot. It would require clear distinctions between modes, mixed configurations, and transitions. It would also require

evidence that different coders can apply those distinctions consistently. The present study therefore records observable change, continuity, difference, and ambiguity without assigning *guṇa* modes.

The wider CTP may eventually combine three layers:

1. **Antahkarana:** the functions and relationships being observed.
2. **Guṇas:** the modes through which those functions appear and change.
3. **Topology:** the overall configuration and its development across an interaction.

This pilot tests the first layer. The other layers remain part of the wider research program.

Appendix C. Work Completed Before and During the Sprint

The Cognitive Topology Project and its use of the *antahkarana* as a reference framework existed before the Digital Minds Research Sprint. Previous work included the general modular approach, the aim of developing a technology-resilient language for cognition, and early conceptual descriptions of the functions.

Preparation before the sprint included literature review, development of the research question, and preparation of the report structure.

The work conducted during the sprint included developing and finalizing the experimental design and protocol, running the formal trials, collecting responses, coding the data, analyzing the results, and evaluating the usefulness and limits of the framework.

LLM Usage Statement

ChatGPT was used as a research, coding, and writing assistant. It helped organize the project, locate candidate literature, test methodological ideas, draft and revise text for clarity, and develop scripts and materials for data collection and analysis. The author developed the Cognitive Topology Project, defined the research direction, made the methodological and coding decisions, ran the study, and reviewed and edited the final report. The author verified the cited sources, checked the dataset and calculations, and takes responsibility for the analysis, claims, and reported results.